



VA-7433

**Integrated M.C.A. (Sem. III)
Examination, December-2023**

IMCA-303

Database Systems

Time Allowed : Three Hours

Maximum Marks : 70

Min. Marks : 28

This question paper contains three sections as under :

Section-A

Max. Marks - 10

This section shall have 01 compulsory question comprising 10 questions (maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

Section-B

Max. Marks - 25

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section-C

Max. Marks - 35

Will carry 35 marks with five long answer type questions comprising one compulsory question (question no. 07) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

SECTION - A

- 1 (i) Define the term Database management system.
- (ii) Define the term meta data.
- (iii) What are null values in DBMS?
- (iv) In employee table {empno, ename, job, Deptno} Department table {Dno, Dname, loc} on the basis of above tables.
Display the employe names with their department names.
- (v) Define the term functional dependency.
- (vi) What are the Data anomalies in DBMS?
- (vii) Define roll back and commit in DBMS.

VA-7433]

2

[Contd...

- (viii) Define the term integrity in DBMS.
- (ix) What is Dead lock in DBMS?
- (x) Define the term Data replica in Distributed Databases management system.

1×10=10

SECTION - B

UNIT - I

- 2 What is Database management system? Explain different levels of Data abstraction in DBMS.

5

OR

- 2 What are entity relationship diagram (ER)? Explain its components and their symbols in detail.

5

UNIT - II

- 3 What is query language? Give details with examples.

5

OR

- 3 Explain the various aggregate function used in SQL with suitable examples.

5

VA-7433]

3

[Contd...

UNIT - III

- 4 What are the Armstrong's Axioms in DBMS? 5

OR

- 4 What is multivalued dependencies in DBMS and how they remove in normalization? 5

UNIT - IV

- 5 What is transaction and their ACID properties in DBMS? 5

OR

- 5 Explain the two phase locking schemes in DBMS. 5

UNIT - V

- 6 Explain the term distributed Database management system and give the advantages of DDBMS. 5

OR

- 6 Explain A-Globe-wait for graphs in DDBMS. 5

VA-7433]

4

[Contd...

SECTION - C

- 7 (a) Design an E-R diagram for the following and also create its related tables :
"A faculty can teach many courses and a course can be taught by many faculty members."

7

- (b) Consider the following relations :
Student (Studid, std_name, year_of_study, basic_stipend, dept_no)
Department (Dept_no, Dept_name, academic_block)
Write SQL queries for the following.
(i) List std_name, year_of_study, Dept_name of all students whose name starts with "k".
(ii) Select names of all the students of Computer Science Dept_name whose basic stipend is more than 8,000 per month.
(iii) Increase the basic stipend of 3rd year engineering dept. by 3,000 per month.
(iv) List the number of students in each department.

8

VA-7433]

5

[Contd...

- 8 What are checkpoints? Discuss the role of checkpoints in Database recovery. Give suitable example in support of your discussion.

10

- 9 Determine the output when the following operations are applied on relation R_1 , R_2 and R_3 given below :

R_1		R_2		R_3
A	B	x	y	A
a ₁	b ₁	a ₁	b ₁	→empty
a ₂	b ₂	a ₇	b ₇	
a ₃	b ₃	a ₂	b ₂	
a ₄	b ₄	a ₄	b ₄	

- (i) Union ($R_1 \cup R_2$).
- (ii) Intersection ($R_1 \cap R_2$).
- (iii) Difference ($R_1 - R_2$).
- (iv) Cartesian cross section ($R_1 \times R_2$).
- (v) Division ($R_1 \div R_2$).

10

- 10 What are concurrent transactions? Briefly discuss the problem encountered by concurrent transactions.

10

- 11 Explain the following terms with suitable examples.

- (a) Lossless Decomposition
- (b) Dependency preserving decomposition.

5×2=10